From Dark Data to Darwin Core: A Hybrid OCR/VLM/Agent System for Metadata Extraction from Mycological Herbarium Labels

Anonymous CVPR submission

Paper ID *****

Abstract

Biological collections hold hundreds of millions of specimens whose label metadata remain largely "dark data." At the Komarov Botanical Institute (LE), the world's fifth-largest herbarium, fewer than 1 % of the mycological holdings are imaged and only ∼22 % carry structured metadata. We present a systematic study of automatic transcription and structuring of herbarium label metadata on a demanding real-world corpus: 394 type specimens of smut fungi (Ustilaginomycotina) from LE, spanning 1873–2009, four languages (Latin, German, English, and Russian including pre-reform orthography), and four text types (typeset, typewritten, handwritten, mixed). Across ***17 experiments*** *we compare (i) a seven-engine classical OCR ensemble with orientation preprocessing, (ii) 18 vision-language models (VLMs) as normalizers in zero-shot, few-shot, and OCR-fusion regimes, (iii) two-stage OCR-VLM+formatter pipelines, and (iv) parameter-efficient low-rank adaptation (LoRA) of Qwen3-VL-8B-Instruct selected by an 83-configuration sweep. All systems are scored against a 21-field Darwin Core-aligned schema with a single unified metric suite. Classical OCR reaches at best* $F_1 = 0.767$*; a zero-shot VLM* 0.807*; five-shot prompting adds* $+6.7$ *points to* 0.874*; and a LoRA adapter trained on only 310 examples in ∼25 min on one GPU attains* $F_1 =$ ***0.972*** *(*$\mathrm{CER} = 0.061$*) on the full corpus—a* $+24.0$*-point gain over the OCR baseline, beating 18 stronger or larger models. We show that merging the adapter and batching inference yields a* $5\times$ *speedup (*2.77 *s/image) at negligible quality loss; that a GBIF taxonomic validator is a quality* signal *rather than an* F_1 *booster; and that a naive LLM corrector* hurts *by substituting modern synonyms for historical names. A four-model consensus audit surfaces 51 likely ground-truth omissions. Finally, applying the production model to 534 unseen* Anthracoidea *specimens yields a* $99.8\,\%$ *parse rate with per-field behavior consistent with the annotated corpus. Our results provide a practical, on-premise recipe for cost-effective mobilization of mycological label data.*

1. Introduction

Biological collections are the material foundation of taxonomy and of one of the great tasks of the life sciences—documenting the Earth's biodiversity. According to Index Herbariorum, the world holds roughly 3 500 active herbaria containing ∼398 million physical specimens [5]. These collections underpin systematic, ecological, climatic, and conservation research and serve as a verifiable record of the presence of taxa in space and time [1]. The Komarov Botanical Institute (LE) holds the largest herbarium in Russia and the fifth largest worldwide, with more than 7.3 million specimens accumulated over 300 years; its mycological fund (∼350 000 specimens) is the largest fungal collection in the country and includes numerous type specimens of Ustilaginales.

The dark-data problem. Despite their scientific value, collection metadata remain "dark data." At LE, only ∼5 % of vascular plants are imaged, the figure for the mycological part is below 1 %, and structured metadata exist for only ∼22 % of specimens. Globally, the rate of metadata transcription has not increased in over a decade and remains the principal bottleneck of biodiversity mobilization into open databases [4]. Manual transcription is expensive, slow, and noisy due to the heterogeneity of curatorial schools and epochs. The resulting data should follow the FAIR principles and biodiversity standards such as Darwin Core [8].

Why mycological type labels are hard. Our working corpus is the smut-fungi (Ustilaginomycotina, "Smuts") subcollection of LE: 394 type-specimen images with inventory prefixes `LE F-*`. A first visual analysis reveals extreme heterogeneity along several axes (Tab. 1): epochs from 1873 to 2009; four languages (Latin scientific names, German, English, and Russian including pre-reform orthography); four text types (typeset, typewritten, handwritten, mixed); diverse date formats; legacy toponyms (e.g. "USSR: Kazachstaniae, Alatau Transiliense") alongside modern DMS coordinates; visually expressed status mark-

Axis	Range / examples
Epochs	1873–2009 (Sydow 1898; Vánky 1988–96)
Languages	Latin, German, English, Russian (pre-reform)
Text type	typeset, typewritten, handwritten, mixed
Dates	`20.10.1898`, `13.VI.1946`, `26.II.1988`
Geography	legacy toponyms; DMS coordinates
Status	Isotypus / Holotypus (colored stickers)
Layout	multi-label sheets, barcodes, envelope spreads

Table 1. Heterogeneity of the smut-fungi (Smuts) subcollection. This diversity is the scientific motivation of our study: it stresses every stage of the pipeline, from orientation to handwriting recognition to taxonomic normalization.

ers (*Isotypus*, *Holotypus* colored stickers); and complex layouts (multi-label sheets, barcodes, envelope spreads). Crucially, none of the closest prior systems is validated on material of such temporal and linguistic breadth, and most target *plant* collections that rely on the World Flora Online rather than fungal nomenclature [6, 7].

Goal. We aim to build and empirically validate a modular pipeline that extracts structured, machine-readable metadata from mycological label images, aligned with a 21-field Darwin Core-compatible schema [8] extended with mycology-specific fields (host plant, type status), with per-field confidence and an audit trail suitable for curator review. Our study is deliberately incremental: each step is compared against the previous one on the same gold set, so that the contribution of every component can be attributed quantitatively.

Contributions.

1. A rigorous, reproducible benchmark of **17 experiments** on a real mycological type corpus, from a seven-engine classical OCR ensemble to modern VLMs, all evaluated with a single unified metric suite (Sec. 4).
2. Evidence that **domain adaptation dominates architecture**: a LoRA adapter (rank 32, 0.5 % trainable parameters) trained on 310 examples raises F_1 from 0.807 (zero-shot VLM) to **0.972** on the full corpus, outscoring 18 stronger or larger models (Sec. 5).
3. A detailed **LoRA hyperparameter study** (83 configurations) quantifying the effect of learning rate, rank, α, epochs, and target modules, with a compact recipe.
4. A practical **production recipe**: adapter merging plus batched inference gives a 5× speedup (2.77 s/image) with no measurable quality loss; we report throughput for corpus-scale deployment.
5. Two carefully designed **negative results**—a taxonomic-API validator is a signal, not an F_1 booster, and a naive LLM corrector degrades quality—plus a **cross-model GT audit** that finds 51 likely annotation omissions.
6. A **generalization test** on 534 unseen *Anthracoidea* specimens, confirming a 99.8 % parse rate and consistent per-field behavior.

2. Related Work

Herbarium digitization pipelines. The closest open system is **Hespi** [6], which chains two-level object detection (sheet components, then fields within the primary label), a text-type classifier, a Tesseract + TrOCR ensemble, cross-checking taxa against World Flora Online and the Australian National Species List, and a final GPT-4o correction. On printed labels it reaches 92.7 % text similarity, on handwritten 88.8 %, and on the heterogeneous DILLEN set 67.1 %; the LLM correction alone adds 6–11 points. Wen *et al*. [7] combine PaddleOCR with a text-only LLM and a three-tier confidence system over 16 Darwin Core fields, reaching 95.4 % field accuracy on *printed plant* labels with fully local inference—an important baseline for whether a multimodal VLM is needed at all. Both target vascular plants and rely on botanical authorities, which are useless for fungi. Earlier work established that OCR on a cropped label is more accurate than on the full sheet, and that Google Cloud Vision achieves only ∼24 % exact matches on Latin names [4], motivating cross-checks with taxonomic references.

OCR/HTR and document understanding. Modern OCR engines include PaddleOCR, Surya, docTR, and Tesseract 5; handwriting recognition traditionally uses TrOCR and, increasingly, general-purpose VLMs. OCR-free models such as Donut and GOT-OCR 2.0 read an image directly to structured text. General and OCR-specialized VLMs (Qwen3-VL, InternVL, Gemma-4, GLM-OCR, and the Datalab family chandra/lift/surya) have rapidly become competitive on document understanding. Our study evaluates representatives of all these families under a single protocol.

Structured extraction, validation, and adaptation. Constrained decoding and schema-guided generation produce syntactically valid JSON. For fungi, the authoritative references are Index Fungorum and MycoBank, with GBIF as a cross-check [2]. We test a GBIF-based validator and an LLM-based corrector, and—unlike prior single-pass pipelines—we compare parameter-efficient fine-tuning (LoRA [3]) against prompting and architecture changes. To our knowledge this is the first end-to-end study on a multilingual mycological *type* corpus of this temporal breadth.

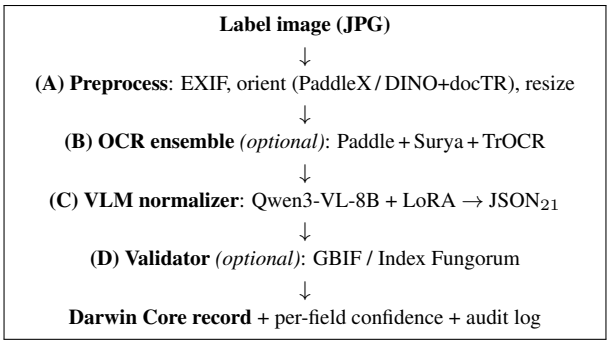


Figure 1. The modular pipeline. The production system uses stages A→C (domain-adapted); stages B and D are studied as ablations (Sec. 4).

3. Method

3.1. Task and 21-field schema

Given a single label image, the system outputs a JSON record with **21 fields** aligned to Darwin Core and extended for mycology, grouped as: *identifiers* (`catalog_number`); *taxonomy* (`scientific_name`, `verbatim_identification`, `type_status`, `determiner`, `verbatim_identification_date`); *collection* (`collector`, `verbatim_collector`, `field_number`, `verbatim_collection_date`); *geography* (`country`, `verbatim_locality`, `verbatim_latitude`, `verbatim_longitude`, `verbatim_elevation`, `verbatim_habitat`); *host/substrate* (`verbatim_substrate`, `organism_remarks`, `identification_remarks`); and *series* (`exsiccata_title`, `exsiccata_number`). Both *verbatim* (as-written) and *normalized* variants are requested for several fields to disentangle reading from normalization.

3.2. Modular four-stage pipeline

The system is modular so that individual modules can be swapped as better models appear (Fig. 1). **(A) Preprocessing / orientation:** EXIF transpose, page-level orientation correction via a document-orientation classifier, a conservative Grounding-DINO + docTR fallback for labels occupying a small image fraction, and resizing to max side $=$ 3000. **(B) OCR ensemble** (optional): classical engines producing tokens with confidences. **(C) VLM normalizer:** a vision-language model consumes the image (and optionally OCR text) and emits the 21-field JSON. **(D) Tool-using validator** (optional): taxonomic and geographic APIs verify normalized fields. Each stage is evaluated independently, letting us attribute quality gains to components. Our final production configuration collapses to stage A followed by a domain-adapted stage C.

3.3. Ground truth, splits, and metrics

The 394 images were manually annotated across all 21 fields (JSONL). For fine-tuning we use a stratified split of **310 train / 79 test** images, with 5 additional stratified images reserved as few-shot exemplars and excluded from both. A single evaluator scores every experiment identically: character/word error rate (CER, WER via jiwer on concatenated fields); *sim*, the RapidFuzz `token_set_ratio`; *hard-R*, recall at a similarity threshold $\tau = 0.85$; *soft-R*, mean per-field similarity where present; F_1-micro (TP/FP/FN summed over all fields, our headline metric) and F_1-macro; *parse_rate*, the fraction of outputs yielding valid JSON; and *sec/img*. Unless noted, "F_1" denotes F_1-micro.

3.4. Fine-tuning setup

We apply LoRA [3] to `Qwen3-VL-8B-Instruct`. Images are capped at max_pixels $= 1024\times28\times28$ ($\approx$900×900 px) to avoid out-of-memory from native-resolution image tokens ($\sim$6 580 tokens otherwise). Training uses bf16, gradient checkpointing, flash-attention 2, effective batch 8, and a cosine schedule with 10 % warmup on a single H100 80 GB. The final recipe, selected by the sweep of Sec. 4, is rank $r = 32$, $\alpha = 256$ ($8\times r$), dropout 0.05, lr $=$ 7e-5, 5 epochs, targeting *all* linear projections (attention and MLP). Training runs in $\sim$25 min; the adapter is 334 MB (43.6 M / 8.81 B $=$ 0.5 % trainable parameters).

4. Experiments

We report 17 experiments organized as a ladder of increasing capability. All use the same 394-image corpus and metric suite; fine-tuning experiments also report the 79-image held-out test split. Hardware is 4–8× H100 80 GB.

4.1. Classical OCR baselines (Exp. 1–4)

We benchmarked seven engines with field extraction by regular expressions only (dates, capitalized tokens after `leg.`/`det.`, etc.), as in historical workflows. Table 2 reports raw (Exp. 1) and orientation-preprocessed (Exp. 2) results. Raw Surya is the best single engine ($F_1 = 0.732$); page-level orientation correction plus resizing lifts it to **0.767** ($+3.5$ points, sim $+4.6$) and speeds up CPU engines 2.5–3.3×. A rise in CER alongside higher F_1 is expected: on de-rotated images engines now read long descriptive bulletins (insertions inflate CER) while the key fields become correct.

An *aggressive* hybrid orientation stage (Grounding-DINO crop by default, Exp. 3) regressed all engines by 1–6 F_1 points, because reading previously upside-down bulletins introduced text absent from the (short) ground truth. A *conservative* hybrid (Exp. 4), triggering the crop path only when the page classifier is unsure (14.5 % of files),

Engine	Exp. 1 (raw)		Exp. 2 (preproc)	
	F_1	sim	F_1	sim
Surya	0.732	0.703	**0.767**	**0.746**
docTR	0.722	0.669	0.721	0.673
PaddleOCR	0.674	0.672	0.731	0.638
TrOCR-printed	0.672	0.669	0.683	0.659
TrOCR-handwr.	0.642	0.617	0.636	0.634
Tesseract 5	0.620	0.627	0.639	0.620
EasyOCR	0.471	0.604	0.478	0.595

Table 2. Classical OCR ensemble (394 images). Orientation preprocessing (Exp. 2) gives the largest single-engine gains to Surya ($+3.5$ F_1) and PaddleOCR ($+5.7$). An aggressive crop-by-default hybrid (Exp. 3) regressed all engines by 1–6 points; a conservative hybrid (Exp. 4) recovered Exp. 2 quality.

Model	mode	parse	F_1	CER	s/img
Qwen3-VL-8B-Instruct	A	0.98	0.807	0.48	16.4
Qwen3-VL-8B-Instruct	B	0.99	0.808	0.50	14.0
Qwen3-VL-30B-A3B-Instr. (BF16)	B	0.99	0.798	0.65	33.0
Qwen3-VL-30B-A3B-Instr. (4-bit)	A	0.93	0.769	0.88	82.4
InternVL3-8B	A	0.99	0.668	0.60	15.4
Qwen3-VL-8B-*Thinking*	A	0.57	0.544	0.74	314.3
Surya (OCR, Exp. 2)	–	–	*0.767*	*1.95*	*1.3*

Table 3. VLM normalizer (Exp. 5). Instruct $\gg$ Thinking; 8B $\geq$ 30B; vision-only (A) $\approx$ OCR-fusion (B). All strong VLMs beat the OCR baseline.

recovered Exp. 2 quality and slightly improved the most independent engine (PaddleOCR, 0.740). We therefore feed the preprocessed corpus (stage A) to all downstream VLM stages. Spearman correlation of per-image similarity consistently selected the triple *PaddleOCR + Surya + TrOCR-printed* as the best OCR ensemble (≈99 % of the oracle upper bound), because PaddleOCR is the most independent engine ($\rho \leq 0.57$ with others).

4.2. VLM normalizer (Exp. 5)

We evaluated 18 models in vision-only (mode A) and vision+OCR-fusion (mode B) settings (Tab. 3). Zero-shot `Qwen3-VL-8B-Instruct` reaches $F_1 = 0.807$ (A) $\approx$ 0.808 (B): *OCR fusion is unnecessary*. Four findings stand out. (i) The 8B *Instruct* model matches or beats the 30B MoE variant (0.807 vs 0.798). (ii) 4-bit quantization of the 30B model regresses by ∼4 points. (iii) *Thinking* variants collapse (parse rate 50–88 %, 15–25 % lower F_1, 15–20× slower), as reasoning consumes the token budget before the JSON block. (iv) Any VLM already beats the best single OCR engine by ∼4 points.

Pipeline	zero-shot F_1	5-shot F_1	Δ
Qwen3-VL-8B A (vision-only)	0.807	**0.874**	+0.067
chandra-ocr-2 md+layout+fmt	0.811	0.869	+0.058
Qwen3-VL-8B B (vision+OCR)	0.808	0.868	+0.060
PaddleOCR-VL-1.6 + fmt	0.803	0.865	+0.062
GLM-OCR (0.9 B) + fmt	–	0.868	–

Table 4. Five-shot prompting (Exp. 6, 5c) adds ∼6 points regardless of whether the pipeline is vision-only or OCR-fusion, JSON-native or text-formatted—indicating the failure mode is task under-specification, not weak vision.

4.3. Two-stage OCR-VLM and few-shot (Exp. 5b–7)

For OCR-VLMs that do not natively emit JSON, we add an 8B *formatter* (image + OCR-markdown → JSON). All "raw" outputs give 0 % parse without the formatter, confirming a two-stage architecture is required for non-instruct OCR models. With the formatter, chandra-ocr-2, PaddleOCR-VL, DeepSeek-OCR, GOT-OCR2 and GLM-OCR cluster at $F_1 = 0.79$–0.87 (Tab. 4); **GLM-OCR** (0.9 B) is the best speed×quality OCR-VLM (0.868 at 19.2 s/img). Adding **five-shot prompting** is the single largest one-step gain in the project: +6.7 points, taking Qwen3-VL mode A to **0.874**, consistently (+5.8 to +6.7) across four pipelines. A separate verbatim "text-gold" benchmark (Exp. 7, 50 hand-transcribed labels) shows that specialized OCR-VLMs transcribe raw text far better (GLM-OCR sim = 0.633) than field-extraction VLMs (Qwen3-VL sim = 0.13–0.17), cleanly separating the two roles: *OCR-VLMs read, VLMs normalize*.

4.4. Foundation-model probes (Exp. 9–11)

To test whether newer/larger backbones help, we ran `Qwythos-9B` and `Qwen3.5-9B-Base` on vision (both $F_1 \leq 0.74$, ∼78 % parse), and benchmarked seven fresh VLMs on the 79-test split (Tab. 5). The best non-LoRA alternative is `lift` (10 B Datalab) + Qwen3-VL formatter at $F_1 = 0.883$—the only fresh model to beat the 5-shot baseline (0.878). Gemma-4-12B (12 B) scores only 0.776; Qwen3.5-9B-Base 0.735; Qwythos-9B 0.711. The lesson is unambiguous: *production-grade instruction alignment and domain fit beat sheer size or novelty*. Notably, surya-ocr-2 (0.7 B, qwen3_5) reaches 0.867—a +13.5 gain over the LSTM-based Surya of Exp. 1.

4.5. LoRA fine-tuning and sweep (Exp. 8, 12–13)

A first LoRA (rank 16, lr 2e-5, 3 epochs) reached $F_1 = 0.920$ on the 79-test, already beating the 5-shot baseline by +4.2 and zero-shot by +10.6 points, while cutting CER from 0.398 to 0.179. An **83-configuration hyperparameter sweep** (∼60 GPU-hours) mapped the landscape

Model (79-test)	size	F_1	CER	s/img
Qwen3-VL-8B + LoRA (Exp. 8)	8 B	**0.920**	0.179	14.9
lift + Qwen3-VL fmt (fs)	10 B	0.883	0.412	37.5
Qwen3-VL-8B A_fs (prod)	8 B	0.878	0.295	16.8
paddleocr-vl-1.6 + fmt fs	1 B	0.875	0.369	31.0
chandra-ocr-2 + fmt fs	4.5 B	0.870	0.422	45.0
surya-ocr-2 + fmt fs	0.7 B	0.867	0.319	23.3
unlimited-ocr + fmt fs	3 B	0.858	0.366	34.0
Gemma-4-E4B-it A_fs	8 B	0.786	0.525	14.7
Gemma-4-12b-it A	12 B	0.776	0.517	13.3
Qwen3.5-9B-Base A	9 B	0.735	0.538	43.1
Qwythos-9B A	9 B	0.711	0.704	37.5

Table 5. Fresh-VLM benchmark on the 79-test split (Exp. 9–11). No model matches LoRA; `lift` is the best non-LoRA alternative. Larger/newer backbones do not help without domain SFT.

Run	r	lr	α	F_1^{79}	F_1^{394}
p5j (prod)	32	7e-5	**256**	0.9564	**0.9720**
p4h	32	**2e-4**	64	**0.9615**	0.9710
p4g-r48	48	1e-4	96	0.9553	0.9694
p4g-r16	**16**	1e-4	32	0.9548	0.9666
lr progression (r=32, 5 ep):					
5e-6→0.83, 2e-5→0.92, 7e-5→0.95, 2e-4→0.96					

Table 6. LoRA sweep highlights (Exp. 12–13). The winner on the small test split ($p4h$) differs from the winner on the full corpus ($p5j$, α=256). Mean F_1 over 82 runs = 0.935 ± 0.024.

(Tab. 6): F_1 rises monotonically with learning rate up to 2e-4 (then overshoots); MLP projections are essential (attn-only loses ∼2 points); rank 16 (167 MB) nearly matches rank 32 (334 MB); and $\alpha = 8{\times}r$ is an underrated lever. Of 82 valid runs, 67 (82 %) exceed $F_1 \geq 0.93$. Evaluating the top-4 adapters on the *full* 394-image corpus (Exp. 13) yields the record F_1 = **0.972** (CER = 0.061, parse = 99.7 %) for r=32, α=256, lr=7e-5, 5 epochs. All four adapters scored *higher* on the full corpus than on the small test split (e.g. +1.56 for the winner), and CER roughly halved (0.104 → 0.061).

4.6. Deployment and validation (Exp. 14–16)

Merging the adapter into base weights preserves F_1 exactly while removing PEFT overhead (1.56×); adding **batched** inference (batch 8) reaches **2.77 s/img**—a 5× speedup over the adapter (Tab. 7)—with F_1 differences within numerical noise (∼1–2 field flips over 294 shared files). External serving (vLLM/SGLang) was blocked by a CUDA-toolkit mismatch and is left as future work. A **GBIF validator** (Exp. 15) matched 85 % of names exactly and caught 10 genuine spelling errors, but the F_1 delta was 0.000: the model is already correct above τ, and canonical forms occasionally conflict with the historical GT convention (blind country normalization, USA → United States, even *hurt* by −0.12). A four-model **GT audit** (Exp. 16) surfaced 51 likely GT omissions via consensus (e.g. missing `type_status`, coordinates, and Latin remarks), while a naive **LLM corrector** for 47 GBIF-flagged names (Exp. 16.1) was net-negative (3 improved, 12 worsened) by substituting modern synonyms (Ustilago→Microbotryum) for historical names.

Method	F_1	s/img	throughput
LoRA adapter (PEFT, single)	0.9720	14.0	4.3 img/min
Merged, single	0.9564*	8.4	7.1 img/min
Merged, batch 8	**0.9716**	**2.77**	**21.7 img/min**

Table 7. Production variants (Exp. 14). *on 79-test. Batch-8 throughput ≈33 GPU-days per 10^6 images on one H100, or ∼8–9 days on 4×H100.

Stage	Best configuration	F_1	CER	s/img
OCR raw	Surya (Exp. 1)	0.732	1.79	2.6
OCR + preproc	Surya (Exp. 2)	0.767	1.95	1.3
VLM zero-shot	Qwen3-VL-8B A (Exp. 5)	0.807	0.48	16.4
OCR-VLM + fmt	GLM-OCR+fmt+fs (Exp. 5c)	0.868	0.51	19.2
VLM 5-shot	Qwen3-VL-8B A_fs (Exp. 6)	0.874	0.30	16.8
Best non-LoRA	lift + fmt fs (Exp. 11)	0.883	0.41	37.5
LoRA (first)	r16 (Exp. 8)†	0.920	0.18	14.9
LoRA (best)	**r32 α256 (Exp. 13)**	**0.972**	**0.061**	14.0
+ merge/batch	merged, batch 8 (Exp. 14)	0.972	0.062	**2.8**

Table 8. Headline progression on the 394-image corpus (except † on the 79-image test split). The cumulative +24.0-point F_1 gain decomposes as preprocessing +3.5, VLM +4.0, five-shot +6.7, LoRA +9.8.

4.7. Generalization to *Anthracoidea* (Exp. 17)

We applied the production model to **534 previously unseen** *Anthracoidea* specimens (23 species), for which no ground truth exists. After identical preprocessing, the model produced valid JSON for **533/534** (99.8 %) images at 12.1 s/img. Manual verification of 15 random labels confirmed near-perfect extraction on printed and typewritten labels; residual errors concentrate on handwritten toponyms and on the normalized `scientific_name`/`determiner` authorities, matching the corpus-level weaknesses (Sec. 5).

5. Results and Analysis

Overall progression. Table 8 summarizes the ladder. The cumulative gain from the OCR baseline (Exp. 1, 0.732) to the final adapter (Exp. 13, 0.972) is +24.0 points. *Fine-tuning on merely 310 examples is the dominant lever*—larger than every architectural change we tried, and larger

Field	n_{gt}	F_1	Field	n_{gt}	F_1
exsiccata_title	322	0.995	verbatim_locality	375	0.961
catalog_number	376	0.993	collector	368	0.955
country	375	0.985	field_number	122	0.954
verbatim_elev.	139	0.982	verbatim_habitat	32	0.867
verbatim_ident.	376	0.975	organism_remarks	11	0.727
scientific_name	376	0.974	identif._remarks	3	0.500
verbatim_substr.	372	0.971	verbatim_ident._date	2	0.000
type_status	360	0.969	determiner	369	0.965

Table 9. Per-field F_1 for the winning adapter (Exp. 13). All four weak fields ($F_1 < 0.90$) have $< 10\,\%$ occurrence; frequent fields are uniformly strong.

than the $+6.7$ prompting gain. The relative error $(1-F_1)$ drops from 0.268 to 0.028, a $9.6\times$ reduction.

Bigger is not better. Across Exp. 5 and 9–11 the same pattern recurs (Tab. 3, 5): a well-aligned 8B instruct model matches or beats 9–30B backbones. The 30B MoE model scored 0.798 vs the 8B's 0.807; Gemma-4-12B 0.776; Qwythos-9B and Qwen3.5-9B-Base 0.71–0.74 with poor ($\sim$78 %) parse rate. Quality is *pre-training* $\times$ *instruction alignment* $\times$ *domain SFT*; "newer" or "larger" alone is insufficient. Qwythos's higher CER (0.704) versus its own base (0.538) shows that reasoning-oriented post-training even *degrades* verbatim transcription.

Per-field quality. For the winning adapter, 16 of 21 fields reach $F_1 \geq 0.97$ (Tab. 9): `catalog_number` 0.993 (barcode+text), `exsiccata_title` 0.995, `country` 0.985, `scientific_name` 0.974. The four weak fields all have $< 10\,\%$ occurrence in the data, so the signal is dominated by sparsity rather than model error. Error categorization shows the dominant modes are *partial* matches (date/locality formatting) and *omissions* on rare fields; *hallucinations* are low ($\leq$ 10 per field). Strikingly, cross-model consensus (Exp. 16) reveals that several apparent hallucinations are in fact *correct* label text missing from our GT (e.g. the Latin remark *"etiam in parte sepulta, haud chlorophylliphera"* on two sheets), implying that fixing the annotation would raise F_1 toward 0.975–0.978 "for free."

Deployment. Merging plus batch-8 inference retains $F_1 \approx 0.972$ while cutting latency to 2.77 s/img (Tab. 7), enabling corpus-scale processing on-premise; the residual -0.04 point on the full run is an artifact of missing records, not a real regression (per-field diffs are $\leq$2 flips/field).

Validation is a signal, not a booster. GBIF matched 85 % of names and detected 10 real spelling errors, yet the F_1 delta was 0.000 because the model is already correct above threshold and canonical forms conflict with the historical GT convention. The naive LLM corrector was net-negative. The practical conclusion is to use taxonomic tools for *per-field confidence and audit*, keeping corrections conservative (genus preserved, sim $\geq$ 0.95, no synonym swaps), and to route only low-confidence records to a curator.

Generalization. On 534 unseen *Anthracoidea* specimens the model achieved 99.8 % parse rate, with fill rates reflecting label structure: near-100 % for core fields (`scientific_name`, `catalog_number`, `country`, `verbatim_locality`) and lower for conditionally present fields (coordinates $\sim$11 %, `type_status` 54 %, exsiccata $\sim$43 %). The model reliably recovers `catalog_number` from barcodes/stamps even though file names carry no inventory number—an important property for scaling to the full $\sim$350 000-specimen collection.

Limitations. Our gold set is modest (394 images) and stratified but small; rare, multilingual, and handwritten fields remain the weakest (Tab. 9), and some GT labels are provably incomplete (Exp. 16). The taxonomic validator does not yet cover historical Soviet/Russian names absent from GBIF/Index Fungorum, and high-throughput serving via vLLM/SGLang is unresolved on our CUDA-12 cluster.

6. Conclusion

We presented a systematic 17-experiment study of automatic metadata extraction from mycological herbarium labels on a uniquely heterogeneous corpus of smut-fungi type specimens. Progressing from a seven-engine classical OCR ensemble ($F_1 = 0.767$) through zero-shot and five-shot VLMs to a parameter-efficient LoRA adapter, we reach $F_1 = 0.972$ (CER $= 0.061$) on the full corpus—a $+24.0$-point gain—using an 8B model fine-tuned on only 310 examples in $\sim$25 minutes on one GPU. Adapter merging and batching yield a $5\times$ inference speedup at equal quality, and the model generalizes to 534 unseen *Anthracoidea* specimens at a 99.8 % parse rate. Two negative results delineate the limits of taxonomic-API and LLM "correction," and a cross-model audit shows the ceiling is partly set by annotation completeness. Future work targets expanded gold annotation for rare fields, a conservative tool-using validator backed by MycoBank and historical gazetteers, and high-throughput serving toward mobilizing the full LE mycological collection.

References

[1] Charles C. Davis. The herbarium of the future. *Trends in Ecology & Evolution*, 38(5):412–423, 2023.

[2] GBIF Secretariat. Gbif backbone taxonomy and occurrence api. https://www.gbif.org/, 2024.

[3] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *ICLR*, 2022.

[4] Daniel Owen, Laurence Livermore, Quentin Groom, et al. Towards a scientific workflow featuring natural language processing for the digitisation of natural history collections. *Research Ideas and Outcomes*, 6:e58030, 2020.

[5] Barbara M. Thiers. Index herbariorum: A global directory of public herbaria and associated staff, 2024. https://sweetgum.nybg.org/science/ih/.

[6] Robert Turnbull, Emily Fitzgerald, Karen Thompson, and Joanne L. Birch. Hespi: A pipeline for automatically detecting information from herbarium specimen sheets. *BioScience*, 2025. arXiv:2410.08740, DOI:10.1093/biosci/biaf042.

[7] J. Wen et al. Automated information extraction from plant specimen labels using ocr and large language models. *Biodiversity Data Journal*, 2025.

[8] John Wieczorek, David Bloom, Robert Guralnick, et al. Darwin core: An evolving community-developed biodiversity data standard. *PLoS ONE*, 7(1):e29715, 2012.